

Drop-Weight Test-Based Evaluation of Nanocomposites Reinforced with Carbon Nanotubes of Varying Geometries

Multifunctional Materials Lab, Mechanical Engineering Department, California Polytechnic State University, San Luis Obispo

Ramanan Sritharan, Luke Granillo, and Wyatt Barnes

SAMPE 2026 Technical Conference and Exhibition, Poster ID # 246



1. Introduction

- Composite materials are widely used in high-performance applications, particularly in the aerospace, automotive, and energy industries.
- These materials boast a remarkable strength-to-weight ratio, but they do have a drawback: poor interlaminar strength.
- To address this limitation, Carbon Nanotubes (CNTs) are employed as reinforcements in traditional brittle composites. By doing so, CNTs enhance the bonding and load transfer between the matrix and the reinforcing material.

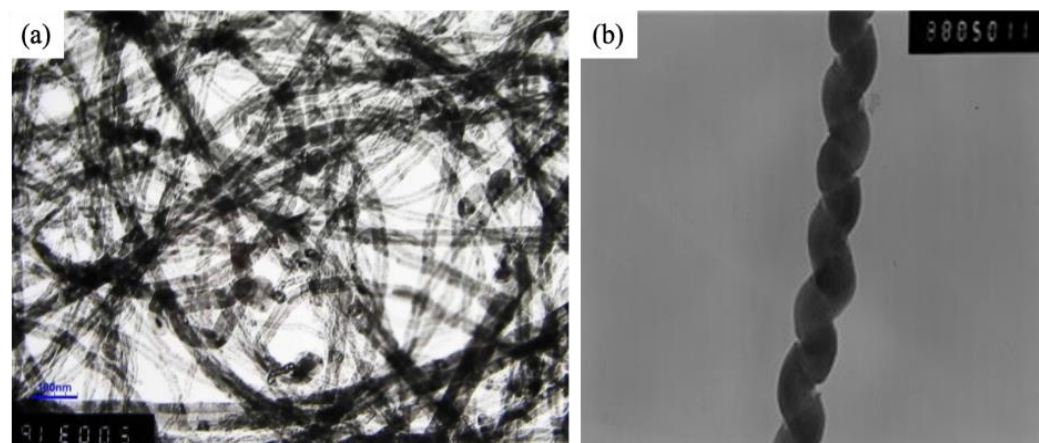


Figure 1: SEM images of (a) multi-walled straight CNT (MWSCNT) and (b) multi-walled helical CNT (MWHCNT).

2. Objectives

- The primary objective of this research study is to evaluate the impact resistance properties of epoxy reinforced with CNTs nanocomposite panels, considering the influence of different CNT geometries (straight and helical), surface chemistry (functionalization), and weight percentages.

3. Fabrication

- Drop weight impact testing was conducted in accordance with ASTM Standard D7136.
- Epoxy panels were fabricated and reinforced with 0.1 wt% CNTs.
- Drop height was calibrated to 5% on specified energy level.
- Failure was evaluated through visual and microscopic analysis.

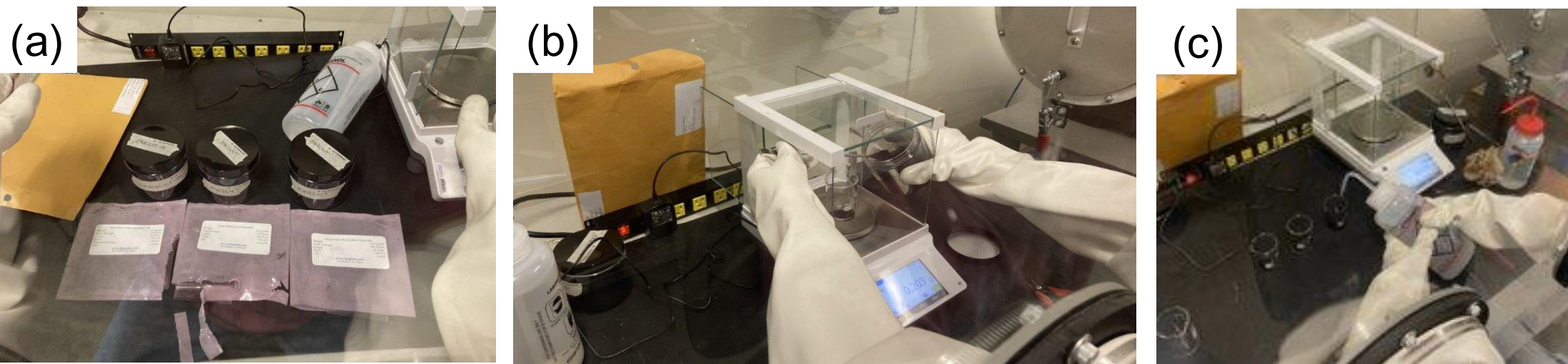


Figure 2: Fabrication steps for processing CNTs; (a) preparing CNTs, (b) measuring CNTs, (c) pouring ethanol into beakers

4. Fabrication (cont'd)

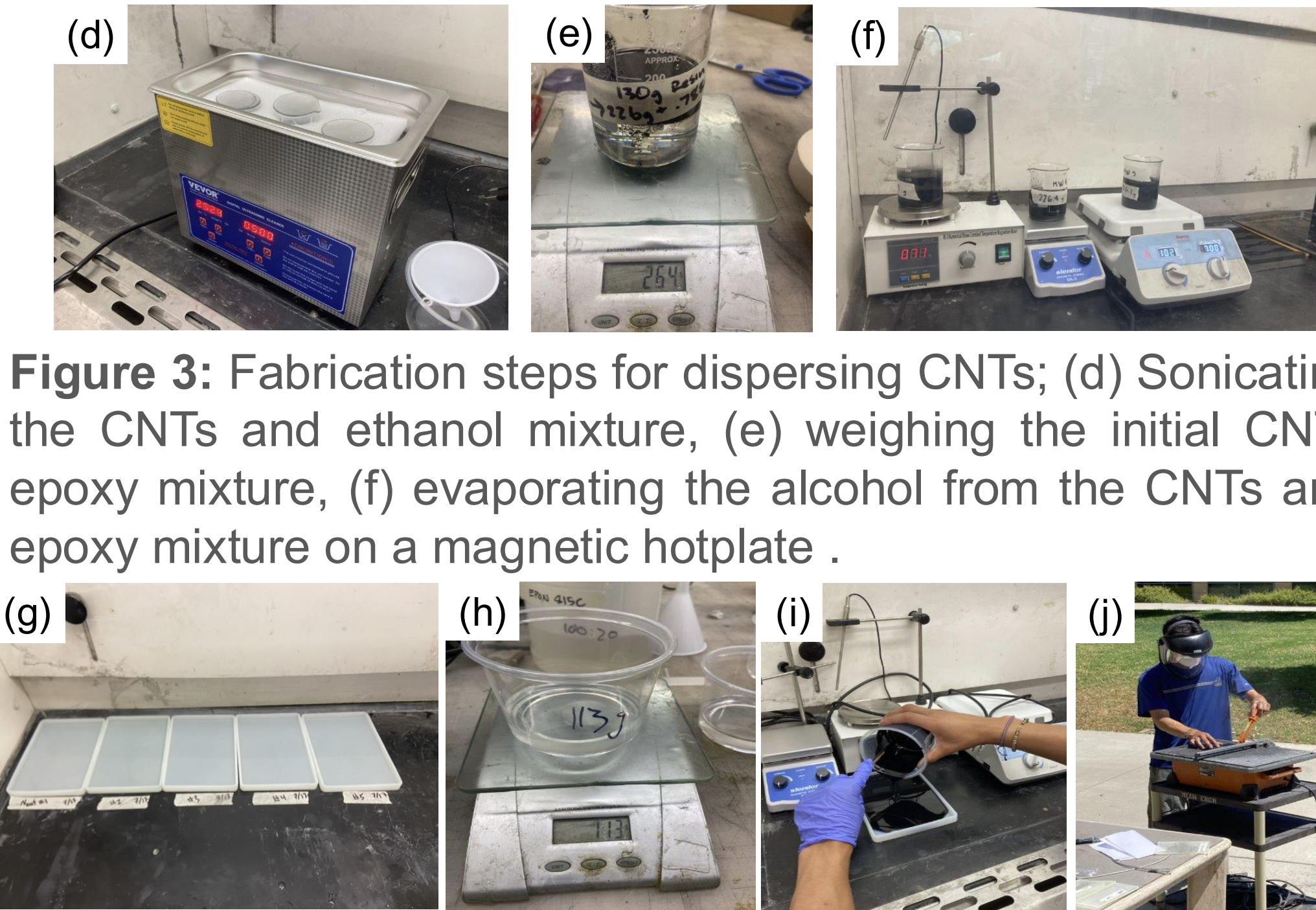


Figure 3: Fabrication steps for dispersing CNTs; (d) Sonicating the CNTs and ethanol mixture, (e) weighing the initial CNTs epoxy mixture, (f) evaporating the alcohol from the CNTs and epoxy mixture on a magnetic hotplate .

Figure 4: Fabrication steps for curing and cutting of test panels; (g) preparing molds for curing (h) preparing hardener, (i) pouring final CNTs-epoxy-hardener mixture, (j) cutting panels to final dimensions per ASTM standard.

5. Testing & Results

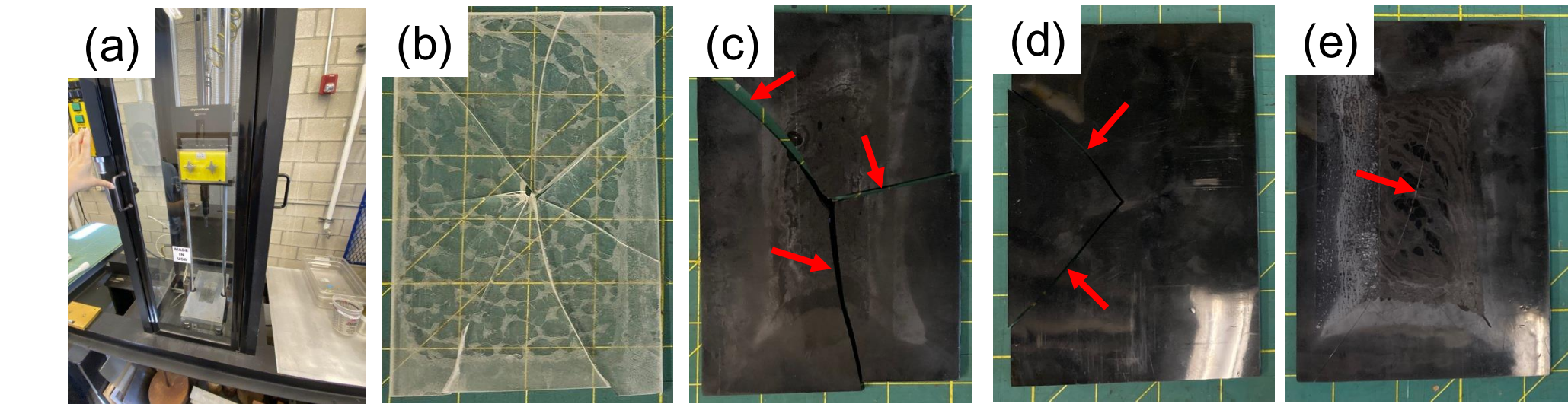


Figure 5: Testing setup and test panels after failure; (a) impact testing equipment, (b) neat epoxy sample #2, (c) MWSCNTs sample #2, (d) MWHCNTs sample #5, (e) FMWSCNTs sample #2.

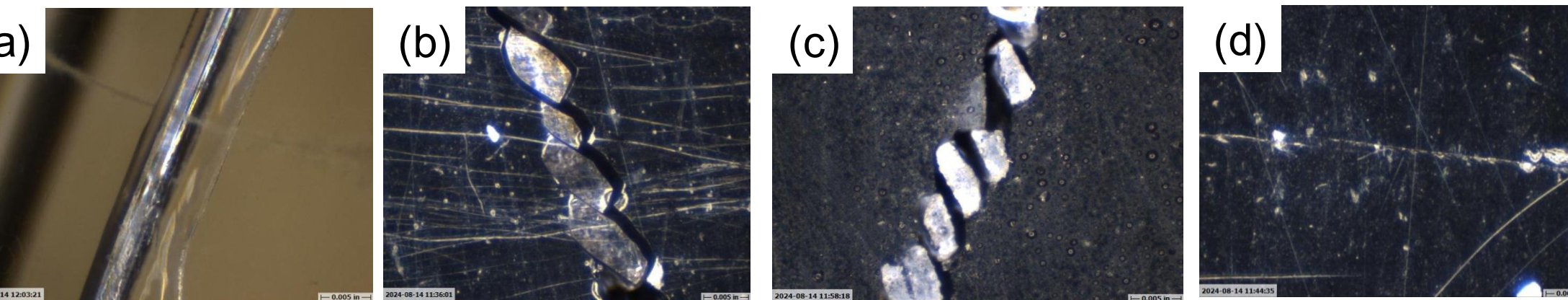


Figure 6: Microscopic images of crack propagations on fractured panels; (a) neat epoxy sample, (b) MWSCNT sample, (c) MWHCNT sample, (d) FMWSCNT sample.

6. Testing & Results(cont'd)

Table 1. Measured Impact Test Parameters and Energy Dissipation.

Samples	Sample Numbers	Impact Energy (J)	Drop Height (cm)
Neat	2	1.6102	4.722
	3	1.6192	4.749
	4	1.6627	4.876
	5	1.6047	4.706
	14	1.8816	5.518
MWSCNT	2	1.7911	5.253
	3	1.7789	5.217
	4	1.7532	5.142
	5	1.7833	5.230
	6	1.7956	5.266
MWHCNT	2	1.7364	5.092
	3	1.7521	5.138
	4	1.7096	5.014
	5	1.7487	5.128
	6	1.7610	5.165
FMWSCNT-OH	1	1.8838	5.525
	2	1.7688	5.187
	3	1.8581	5.449
	4	1.7498	5.132
	5	1.7487	5.128

7. Conclusion

- Overall, CNT reinforcement improves the impact resistance of epoxy. FMWSCNT-OH samples showed the least damage and lowest failure rate while MWSCNT and MWHCNT showed moderate improvement. CNT functionalization is key to enhancing impact performance in polymer composites.

8. References

- R. Sritharan, "Enhancing the short-beam strength of composite laminates using helical carbon nanotubes", Composites Part B, 2021.
- R. Sritharan, "Improvement of the tensile properties of laminated composites reinforced with Heli-Coil forms of carbon nanotubes", Polymer Composites, 2023.

9. Acknowledgment

- This work was supported by the Constant and Dorothy Chrones Endowed Professorship, the Morris P. Taylor and Joy O. Richardson Mechanical Engineering Laboratory Support Fund, and the College of Engineering Summer Undergraduate Research Program (SURP).